
DALIA ALY

SENIOR SOFTWARE DEVELOPER

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WEBSITES

- GITHUB: <https://github.com/daliaalymohamed>
- Portfolio: https://daliaalymohamed.github.io/my_portfolio/

SUMMARY

Senior Software Developer with over 10 years of experience in designing, developing, and maintaining innovative software solutions. Expert in full-stack web development, systems design, and project management. Recent focus on applied AI: LLM integration and automation pipelines. Most recently, built and shipped a commercial inventory management system with enterprise SSO, two-factor authentication, role-based access control and SOC 2-oriented security hardening. Thrives in collaborative environments to deliver exceptional results and continuously learn new technologies. Committed to improving processes and creating user-centered applications.

SKILLS

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|--|---|
| ○ React.js development | ○ Node.js |
| ○ Next.js (App Router) & TypeScript (strict) | ○ GraphQL query optimization |
| ○ JWT authentication (sessions & invalidation patterns) | ○ Open Object Framework “Odoo” |
| ○ Proficient in Passport.js | ○ Proficient in Express.js |
| ○ Containerization & deployment (cloud and self-hosted) | ○ PostgreSQL database management |
| ○ Database management with MongoDB / Mongoose (indexes, aggregation) | ○ Redux state management |
| ○ RESTful API development | ○ MySQL database management |
| ○ Version control expertise "Git" | ○ UI development with Material-UI |
| ○ Responsive styling with Tailwind CSS | ○ Responsive UI with Bootstrap |
| ○ LLM Integration & AI-powered development | ○ Caching & in-memory data management (Redis) |
| ○ Prompt Engineering & Optimization | ○ Full-text search integration (MeiliSearch) |
| ○ AI pipeline development & web | |

scraping automation

- OAuth 2.0 / OpenID Connect SSO (PKCE; Okta and Entra ID compatible)
- TOTP two-factor authentication
- Security hardening & SOC 2 readiness (CSP, rate limiting, encryption at rest, audit logging)
- Automated unit & integration testing

- Spec-Driven Development
- System Architecture & Database Design (ERD, schema)
- Role-Based Access Control (RBAC) design
- Secure authentication flows (magic link, token hashing, bcrypt)
- API & notification system architecture (contextual recipient resolution, RBAC-based routing)
- GitHub Actions (CI)
- DevSecOps (gitleaks, Semgrep, npm audit, Dependabot)
- Cloud file storage (presigned URLs)
- Real-time notifications (Server-Sent Events) & transactional email
- TanStack Query, React Hook Form & Zod validation
- Reporting & exports (Excel, PDF, QR code labels)

TOOLS

AI Tools & Platforms

- OpenAI API (GPT, TTS)
- OpenRouter (multi-model AI routing)
- Hugging Face (models & inference APIs)
- Axios / Fetch (HTTP client for API integration)

System Design & Documentation Tools

- Claude AI (claude.ai web, Claude Code CLI, VS Code extension)
- MCP (Model Context Protocol) — connector integration for AI workflows
- ERD & sequence diagram authoring (Mermaid, drawio-app)
- Google Stitch (UI/screen design prototyping)

Testing, DevOps & Integrations

- Vitest · Testing Library · ESLint
- Docker / docker-compose
- Vercel · PM2 · Nginx
- Keycloak (SSO / OpenID Connect identity provider)
- AWS S3 · SendGrid · pino (structured logging) · node-cron

WORK HISTORY

Senior Software Developer, Aug 2014 – June 2025

Central Department For Digital Community Development – New Administrative Capital

- Led end-to-end software development life cycle by gathering user requirements, designing scalable architectures, and delivering tailored solutions that improved system usability by 30%.
- Identified performance bottlenecks in legacy systems and recommended strategic upgrades, resulting in a 25% improvement in application speed and maintainability.
- Architected modular full-stack applications, ensuring seamless integration between frontend, backend, and database components.
- Conducted regular code reviews and implemented testing, reducing post-deployment bugs by 40% and increasing system uptime.
- Authored comprehensive technical documentation and managed Git-based source control workflows to ensure traceability, scalability, and ease of future enhancements.
- Partnered closely with product managers, UX designers, and DevOps engineers to deliver high-quality features on time and within scope using Agile methodologies.
- Developed and optimized high-traffic government websites using React.js and Node.js, ensuring 99% uptime and optimal user experience across devices.
- Revamped and enhanced the Kenanaonline portal using modern CMS tools, increasing monthly traffic engagement by 35%.
- Evaluated and integrated emerging technologies into production environments, improving API performance and deployment efficiency.
- Engineered custom ERP modules using Odoo, streamlining business operations and reducing manual data entry by 50%.
- Diagnosed and resolved critical production bugs in real-time, ensuring minimal downtime and maintaining user trust on public-facing platforms.
- Enforced strict adherence to W3C standards and accessibility guidelines, ensuring cross-browser compatibility and consistent user experience.
- Optimized websites for SEO best practices and cross-browser support, boosting organic search visibility and broadening audience reach.
- Acted as a key liaison between development teams and department leadership, providing regular updates and aligning technical strategy with organizational goals.
- Tracked and reported progress on 30+ active tasks using Jira, ensuring timely delivery and stakeholder transparency.
- Stayed current with evolving tech trends in full-stack development and contributed to internal knowledge bases, improving team-wide understanding and adoption of new tools.
- Proactively supported ad hoc requests, including mentoring junior developers and contributing to cross-team initiatives.
- Integrated AI-driven features into applications, including automated content generation and intelligent data processing workflows.

EDUCATION

9-Month Diploma: Open Source Department, Sep 2013 – Jun 2014

Information Technology Institute - Smart Village

KNOWAY course (Make a choice, Goal setting & Winning others sessions)

Bachelor of Commerce: Business Information System, Sep 2008 – Jun 2012
Helwan University - Giza, GZ

- Graduation Project Grade: Excellent
- Last Year Grade: Excellent , GPA: Excellent

High School: Sep 1995 – Jun 2008
Aziz Abaza Language School - Cairo, C

ACCOMPLISHMENTS

Inventory Management System (Full-Stack Web App) 2026

Role: Full-Stack Developer

Tech Stack: Next.js 16, React 19, TypeScript, MongoDB / Mongoose, Redux Toolkit, TanStack Query, Tailwind CSS, shadcn/ui, Keycloak (OIDC), AWS S3, SendGrid, Vitest, GitHub Actions, Docker

- Built and shipped a commercial, source-available asset-tracking system covering the full item lifecycle (intake, assignment, return, retirement) with QR-code tracking, audit logging and reporting.
- Designed a configurable role-based access control system (resource × action permissions), enforced on the server for every API route, not just in the UI.
- Built enterprise authentication: SSO via OpenID Connect with automatic account creation and an emergency admin login, mandatory TOTP two-factor authentication with recovery codes, JWT sessions in HttpOnly cookies, account lockout, a 90-day password-expiry policy and admin-approved password resets.
- Hardened the application towards SOC 2 readiness: rate limiting, CORS and security headers, field-level encryption of personal data, an append-only audit log with external forwarding, Zod input validation and dependency vulnerability fixes.
- Implemented approval workflows (request → approve/reject → reassign → return) with full history, plus real-time notifications over Server-Sent Events and email delivery through SendGrid.
- Built 7 filterable, paginated reports exportable to Excel, PDF generation and printable QR labels, plus a live dashboard.
- Wrote 2,000+ automated tests (Vitest + Testing Library).
- Built a GitHub Actions CI pipeline that gates every push and pull request on linting, type checking, 2,000+ tests, a production build, dependency vulnerability audits, secret scanning (gitleaks) and security code scanning (Semgrep), with actions pinned to commit SHAs and Dependabot keeping dependencies updated.
- Wrote deployment runbooks for Vercel, Docker and self-hosted (PM2 + Nginx), plus data migration and rollback scripts for switching the login method.
- Planned and specified the system before building it: wrote 33 feature specs, tracked work through GitHub Issues and pull requests, and used AI-assisted development with Claude Code.

AI-Powered Brochure Generator 2026

Role: Developer

Tech Stack: Python, OpenAI API, OpenRouter, Web Scraping

- Developed an AI-powered system that generates dynamic brochures using web scraping and LLM-based content generation.

- Built automated pipelines to extract, process, and transform website data into structured marketing content.
- Optimized API usage for cost efficiency and performance across multiple AI providers.

AI-Chat Bot Assistant 2026

Role: Developer

Tech Stack: Python, OpenAI API, Ollama

- Developed a streamlining chat-bot

AI-Airline Assistant 2026

Role: Developer

Tech Stack: Python, OpenAI API, OpenRouter

- Integrated text generation, image generation, and text-to-speech (TTS) to create rich multimedia outputs.

Telemid Ticketing System 2024

Role: Supervisor

Tech Stack: React.js, Redux, Node.js, MongoDB, Express.js, JWT, Material-UI, Docker

- Led the development of a containerized ticketing platform for internal use.
- Applied Docker for scalable deployment and efficient environment management.

Ta2heel Web & Mobile Application 2021 – 2025

Role: Developer

Tech Stack: React.js, React Native, Node.js, MongoDB, JWT, websockets, Bootstrap

- Built accessible web and mobile platforms to support national employment services.
- Implemented real-time updates using WebSockets.

Kenanaonline PWA Portals 2021 – 2022

Role: Developer

Tech Stack: React.js

- Maintains and updates the front-end of Kenanaonline portals using React.js.
- Ensures ongoing functionality and user interface consistency for high-traffic content delivery.

GTA Portal – Giza Tourism Administration Jun 2016 – Jul 2016

Role: Developer

Tech Stack: Open Object Framework, PostgreSQL

- Developed and customized tourism administration modules using Odoo 8.

Payment System – “PSU” 2015 - 2016

Role: Developer

Tech Stack: Open Object Framework, PostgreSQL

- Implemented a payment and accounting module using Odoo 8.